

Requirement	ID	Requirement Origin	Test Case	Status	Comments
Create a new input field for users.	#1	Customer	Initiate action by creating specific language.	Success.	Improve interaction.
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A simple RTM

10Pearls Session Notes

testing is to detect the defects in the product and resolve them

Quality is to create customer value by fulfilling the customer requirements.

QA is a proactive approach to prevent defects in the application. (from start to end of the SDLC) (process-oriented) (involves all the teams)

QC is a reactive approach where defects are identified and resolved (product-oriented) (conducted before the delivery of the product to the client) (including walkthroughs, inspections, reviews, and testing) (involves testing team only)

Testing environments:

QA: mimics production environment configurations, used to test the system and smaller in scale. (functional, performance, and security testing)

Staging: resembles production environment (testing before deployment)

UAT: represents real-world scenarios where end users and stakeholders can test the product.

Production: a live operational system for end users (final stable version of the software) scaled for high performance and availability.

QC Activities:

Walkthroughs: an informal step-by-step examination of a code or document by stakeholders led by the creator. (early defect detection and clarification of requirements)

Inspections: Formal and structured examination of documents or code by a specific team focused on defect detection and prevention and quality assurance (deviation from quality standards) led by a moderator.

Reviews: systematic examination of document or code (less formal than inspections) for defect detection, prevention, quality, and improvement of the overall deliverable quality (peer reviews, code reviews, design reviews)

Testing: executing system and identifying defects, validating functionalities, and ensuring that they meet the customer requirement (verification and validation)

CI/CD Pipelines(continuous integration and deployment): Automate the process of source, build, test, and deploy (includes automated integration, testing, and deployment of the system)

STLC (Software Testing Lifecycle)

requirement grooming: the quality assurance team analyzes the requirements and if anything is missing or not clear they tackle that by contacting stakeholders.

Test planning: Plan the scope, objectives, testing strategy, test cases, test data, environment, resources requirement, time, cost, test deliverables, and milestones, and assignment of roles and responsibilities. (produces test plan)

Test cases: writing test cases, test data, scenarios, expected results reviewing and validating test cases.

Test environment setup: By the developer or client (an environment where all the test cases will be executed)

Test execution: Test cases are executed in the environment, defects are logged, and again those defects are fixed. (Repeat until all the defects are resolved) (test reports and defect logs)

Test closure: ensure all the test-related activities are completed and the product is ready to deliver. (test summary report, feedback, improvements, defect tracking, environment clean-up, and test closure report)

RTM: mapping and tracing requirements, test cases, test results, and test issues.

automated testing: precision, scalability, time, efficiency. Mostly used for regression testing and parallel execution of test cases on multiple machines.

Performance testing refers to the evaluation of overall system performance under different conditions.

Load testing refers to determining the maximum limit of load on which the system performance is not affected.